

lect 1

Date -

lect 1

Myth - Randomness ^{may} → Exist or doesn't Exist

Random process exists forever, its outcome is unpredictable

→ The value the signal takes at a next instance can't be accurately predictable

Challenges in time-series analysis

Randomness means that there is no precise mathematical fn to describe that process.

Randomness & Uncertainty

1. Lack of a precise mathematical fn to desc. the process of interest (leading to a probabilistic framework)

2. To be able to draw inferences on the ensembles from a single realization.

3. Estimation of "unknowns" from (uncertain) observations / knowns.

→ predominantly deterministic process

In general any process ^{will be} characterized by some uncertainty

↓ Question

To what extent?

There are class¹ of random processes which can't be predicted at all.

↳ Given any amount of history of the random process, you can't actually make an improvement average.

↳ Ideal Random process

↳ no scope for prediction

} Time series

→ deterministic trend

↳ some scope of prediction.

→ Best prediction for any random process is Average,
↳ no historical data.

Φ. Sources of uncertainty and randomness

1. Uncertainty in process knowledge:

→ process characteristics are seldom known accurately and/or completely. eg. atmospheric process, roll of a die, grade of a student etc.

→ From a prediction point of view, therefore, only outcomes with chances can be started at best.

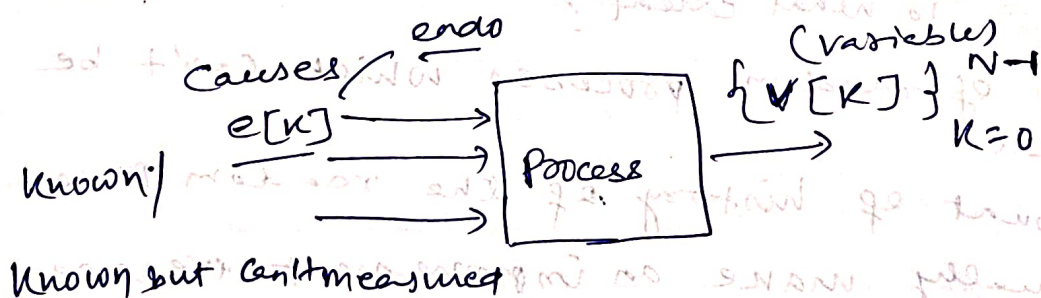
2. Uncertainty in measurements:-

→ Every sensor introduces its characteristics into measurements.
Hence, in reality

$$\text{Observation} = \text{Truth} + \text{perception (of the sensor)}$$

→ Sensor characteristics are usually not fully understood and therefore have to be treated as random variables.

3. Unknown Causes: In several situations, the Causal variables are ~~to~~ unknown and/or known, but cannot be measured ~~and~~ quantified.



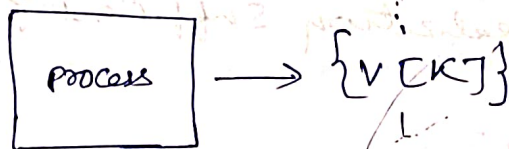
- Build Model from Data ——— univariate
- ①
 - ② Not gopt to take Causes into account
- Multivariate
- ③ — Causes into a/c

Date - 12/09/25

Leet-2

$\square \rightarrow$ discrete
 $\circ \rightarrow$ continuous

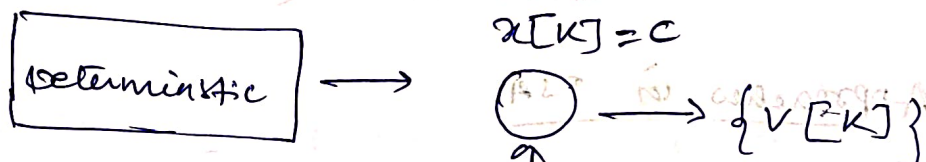
Prof. Babji
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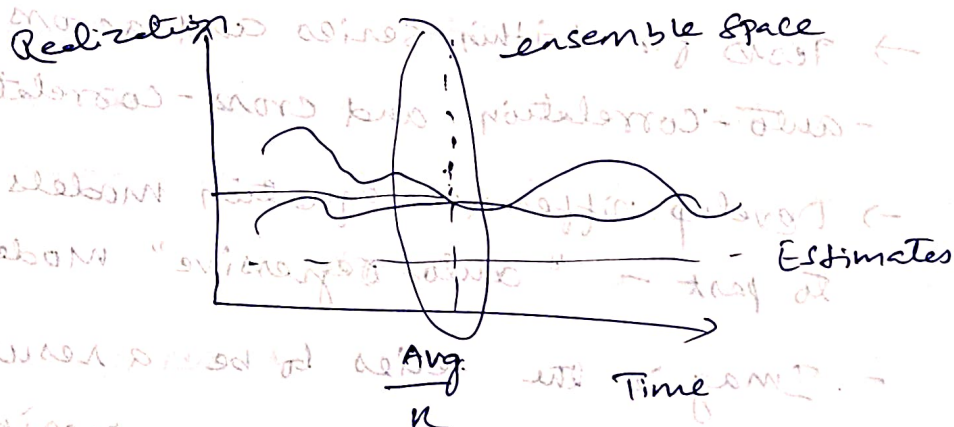
The "Realization" challenge

A time-series is an ordered collection of random variables. Therefore, a single time-series is one of the many possible combinations.

- A single time-series is said to be a realization of the random process.



- The collection of all possible realizations is called ensembles.



Time series Analysis is all about giving you a tool for taking into acc such uncertainties in your record and also how uncertainties in data is going to imp propagate uncertainties in inferences.

Q1. How good are the estimates that I draw from a single realization? ~~that~~

Q2. Is ~~a~~ a single realization sufficient?

A unifying Question -

— Under what conditions can the inferences drawn from a single record of data be meaningful and useful?

1. Stationarity: - Invariance property of the process w.r.t time (or space)
2. Ergodicity: Ability to replace ensemble averages with time (or spatial) averages.

Approaches in TSA

Approaches depend on objective!.

1. Forecasting: — The most prevalent objective of TSA — applies to every field.

→ Tests for within-series and across-series predictability

- auto-correlation and cross-correlation functions

→ Develop difference equation models relating present to past — "auto-regressive" models.

- Imagine the series to be a result of an unpredictable shock wave passing through a filter — "moving average" models.

— Build mixed models that can also include other effects such as trends, random walk, non-stationaries, etc. (seasonal) ARIMA models.

2. Notation of periodicities:-

A common goal of many engineering, meteorological, astronomical and biomedical applications.

- Frequency-domain or spectral representations - extensive use of Fourier transform (other transforms are also used in advance analysis)

→ Notion of a "random periodic process"

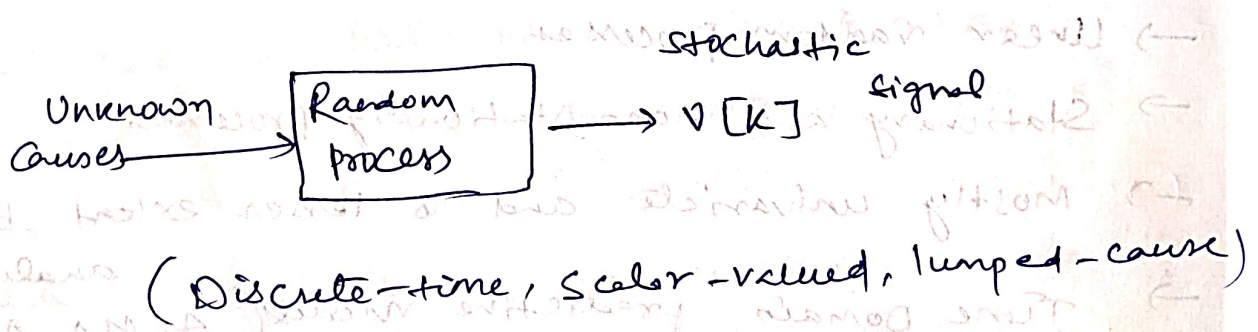
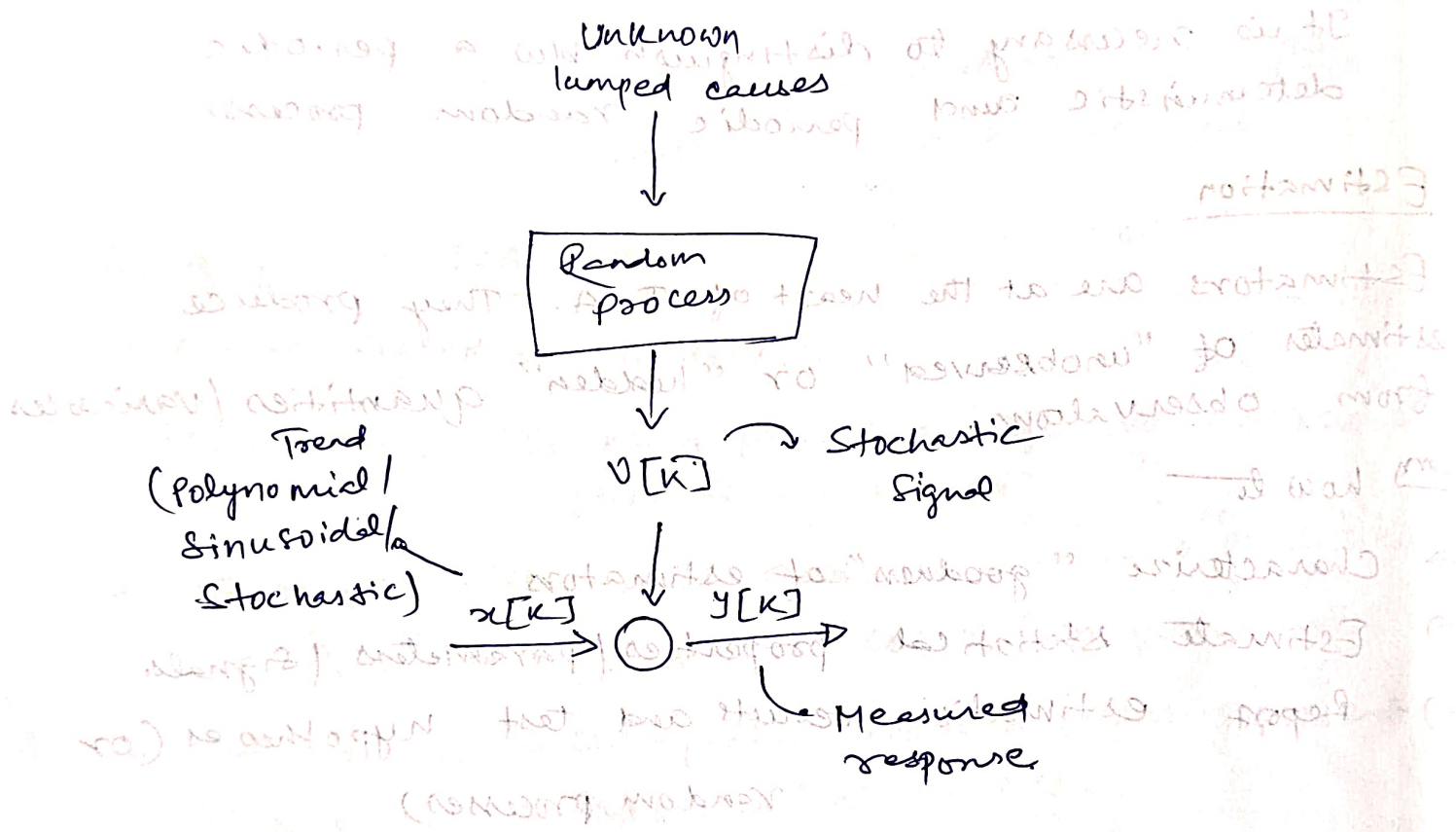
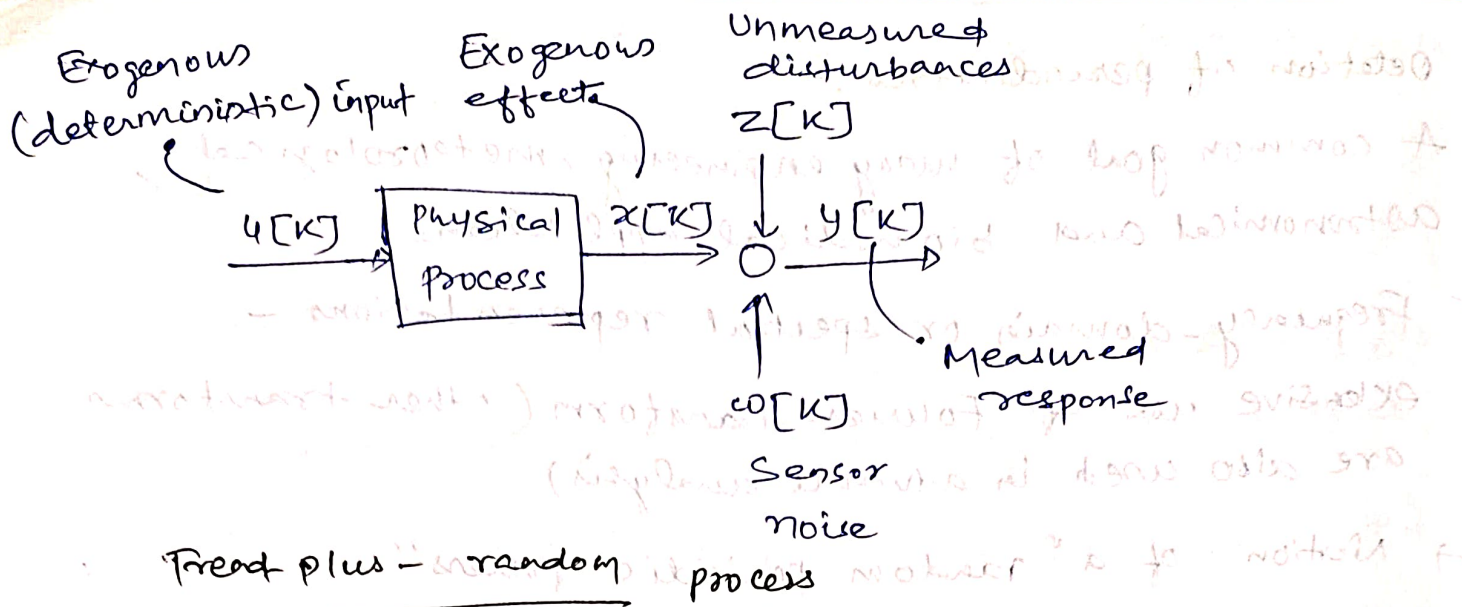
It is necessary to distinguish b/w a periodic deterministic and periodic random process.

Estimation

Estimators are at the heart of TSA. They produce estimates of "unobserved" or "hidden" quantities (variables) from observations.

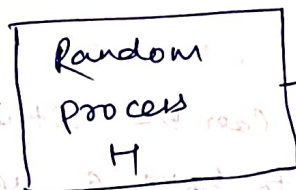
learn how to

- Characterize "goodness" of estimators
- Estimate statistical properties / parameters (signals)
- Report estimation results and test hypotheses (or random processes)
 - Linear random processes.
 - Stationary and Non-Stationary processes
 - Mostly univariate and to lesser extent, bivariate analysis.
 - Time domain predictive models (ARMA, ARIMA, SARIMA)
 - Frequency domain (spectral) analysis (deterministic and stochastic)
 - Estimation theory (MM, LS, MLE and Bayesian estimator or their properties)



White noise
(endogenous)
 $e[k]$

Abstract
(unobserved)



Stochastic
Signal

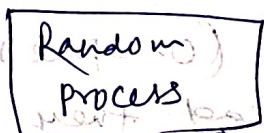
$v[k]$

Real
(observed)

(Discrete-time, scalar-valued, endogenously driven)

Bivariate random process

Observed Input
(random signal)
 $v_1[k]$



unknown
lumped causes
 $w[k]$

Observed output
(random signal)
 $v_2[k]$

(Discrete-time, bi-variate, scalar-valued,
externally plus endogenously driven)

The cornerstone of theory of random processes is the concept of a random variable and the associated probability theory.

Random Variable

Defⁿ: - A random variable (RV) is one whose value set contains at least two elements, i.e. it draws one value from at least two possibilities. The space of possible values is known as the outcome space or sample space.

Examples: Toss of a coin, roll of a dice, outcome of a game, atmospheric temperature

$$\{ \dots, v[-1], v[0], v[1], v[2], \dots \}$$

Formal Definition

Outcomes of random phenomena can be either qualitative and/or quantitative. In order to have a unified mathematical treatment, RVs are defined to be quantitative.

Defⁿ (Priestley (1981))

A RV X is a mapping from the sample space $\mathcal{S}$ onto the real line st to each element $s \in \mathcal{S}$ there corresponds a unique real number.

➤ In the study of RVs, the time (or space) dimension doesn't come into picture. Instead they are analysed only in the outcome space.

Two broad classes of RVs

- When the set of possibilities contains a single element, the randomness vanishes to give rise to a deterministic variable.
- Two classes of random variable exist!

Discrete-valued RV: discrete set of possibilities
(eg. roll of a dice)

Continuous-valued RV: Continuous-valued RV
(eg. ambient temperature)

Focus of this course: Continuous-valued random variables
(with occasional digression to discrete-valued RVs)

Do random variables actually exist?

Randomness — is given to any variable or signal which is not accurately predictable, i.e. the outcome of the associated event is not predictable with zero error.

In reality, there is no reason to believe that the true process behaves in a "random" manner, it is merely that since we are unable to predict its course, i.e. due to lack of sufficient understanding or knowledge that any process becomes random.

Randomness is, therefore, not a characteristic of a process, but is rather a reflection of our (lack of) knowledge and understanding of that process.

Probability Distribution

The natural recourse to dealing with uncertainties is to list all possible outcomes and assign a chance to each of those outcomes.

Examples:

➤ Rainfall in a region: $R = \{0, 1\}$, $P = \{0.3, 0.7\}$

• face value from the roll of a die: $R = \{1, 2, \dots, 6\}$,
 $P(\omega) = \{1/6\} \forall \omega \in R$

The specification of the outcomes and the associated probabilities through what is known as probability distribution completely characterizes the random variable.

Probability Distribution Functions

Probability distribution function

Also known as the cumulative distribution function,

$$F(x) = \Pr(X \leq x)$$

Probability distribution functions can be either continuous or piecewise-continuous (step-like) depending on whether the RV is continuous or discrete valued, respectively.

They are known either a priori (through physics or postulates) or by means of experiments.

When the density function exists i.e. for continuous-valued RVs,

1. The density function is such that the area under the curve gives the probability,

$$\Pr(a < x < b) = \int_a^b f(x) dx$$

$$\Downarrow$$
$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad (1)$$

2. The density function is the derivative (w.r.t x) of the distribution function.

$$f(x) = \frac{dF(x)}{dx}$$

For discrete-valued RVs, a probability mass function (p.m.f) is used.

Density functions

1. Gaussian density fn $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$
2. Uniform density fn - $f(x) = \frac{1}{b-a}, a \leq x \leq b$
3. Chi-Square density fn $f_n(x) = \frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2-1} e^{-x/2}$

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Lect-1 : 11TM

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Time Series

Dependent - evolving as a f^n of independent variable.

Independent

Eg:- Temperature $\rightarrow$ evolving as a f^n of time

Space $\rightarrow$ Image
Pixel

Real. life Eg: Stock Market
Price of stocks
 f^n of time

Eg: Revenue of a company

Time Series - Periodicity / oscillation

- Fluctuations

- Combinations of both

Trends

Combine two or more - Get Time series.

Seasonality $\rightarrow$ Q3 to Q3 $\rightarrow$ year on year
Analysis 24 25

Fluctuations in the Data

- Modelable ?

- Random (Not Modelable) ?

Heart Rate $\rightarrow$ Completely Random? $\rightarrow$ No

Modelable $\checkmark$

Not all Fluctuations are Completely Random,

Not Modelable $\rightarrow$ Can't make Educated Guess
Unpredictable about the process.

Eg: There hardly any process for which we can't
predict. Every process has some relationship
in the past (historical).

That's why Data Science is an important.

Dice — Not an important process.
Coin toss —

Coin toss in cricket match $\rightarrow$ it does not have influence from
the past.

$\hookrightarrow$ Not Modelable

$\hookrightarrow$ Sequence of independent events
identical events.

IID — Independent and Identically
Distributed

$\hookrightarrow$ Bernoulli

Not Modelable $\rightarrow$ independent of past.

Rock Baby —
Sleep

rock is different patterns,
Brains get difficult to model.

Fan noise —

Sleep

Completely Random
that your Brain gives up

patient — Brain interested.

Meditation $\rightarrow$ Calm waves

gamma to Delta waves

insured to sleeping

high frequency
alert β

Deep meditation

Q. 1, 7, 3, 15, ?

✓ (gamma)

6.5

x_i is the data $i = 1$ to 4

$\hat{x}$ is my prediction

$$\text{Obj J} = \frac{1}{N} \sum_{i=1}^n (x_i - \hat{x})^2$$

$$\frac{ABS}{\sum_i |x_i - \hat{x}|}$$

error $\pm$ Cancellation

↳ Sq. root

→ Mode.

$$\text{Min } J = \frac{1}{N} \sum (x - \hat{x})^2$$

Derivative

upwards only - Dept of

No Charge

perivalative should zero

change of service ~~to~~

~~Chopra~~ Positive

Derivative should zero

Rate of change of derivative < 0

negative

$$\min J = \frac{1}{N} \sum_i (x_i - \hat{x})^2$$

$$\frac{\partial J}{\partial \hat{x}} = 0$$

$$\frac{1}{N} \sum_i (x_i - \hat{x}) (-1) = 0$$

$$\frac{1}{N} \sum_i x_i = \frac{1}{N} \sum_i \hat{x}$$

$$\hat{x} = \frac{1}{N} \sum_i x_i \quad \text{--- Mean}$$

Mean is the best predictor if you want to minimize the Squared Error.

If you don't have any other info, avg. is the best no. for prediction.

If we ^{want to minimize} absolute error, Median is the best predictor.

probability and statistics — connecting with prediction

In nature all processes are related with its past, maybe a fn of space / time but it is time series

Matrix $\leftarrow$ Image is a fn of space. 2D

~~Matrix~~ $\leftarrow$ Video — Space & time 3D
Tensors $\leftarrow$ Temperature — 1D — time
 $\leftarrow$ Vector

Course 1D - time series
Temperature, pressure etc.

And linear relationship

Quiz 1	25	paper Based } one easy other hard
Quiz 2	25	
End Sem	50	Moderate
Project		split up not revealed
100		
Attendance	→	-10
Assignment	→	0
Ans	→	30-40
Web	→	30-40
3 yrs	→	Top - 60-65
chelle M Test	→	

Answer a Q? no one gives ✓ high mark.

tough Q? → liberat in for

Relative Sing shot correction

Make thiopk in Exam

no direct Q? from anywhere.

Book! prob & statistics by Montgomery

Date 12/09/25

Lect - -

What is a time-series?

Time-series is a set of data observed (or measured) at regularly or irregularly spaced time instants.

or a fn of any independent variable

(except time etc)
length, frequency etc.

→ sampled-data

→ consists of processing the observed data to extract meaningful and useful information.

→ used for various purposes. such as forecasting (prediction) control, pattern recognition, fault detection etc.

▶ Classical Statistical Inference procedures $\Rightarrow$ Independent observations are obtained under identical conditions (Independent & Identical Distribution) IID

▶ Time series data — Sequence of observations recorded at specified time-intervals such as hours, month, or years etc. over a period of time.

▶ Time series data at adjacent points are dependent

Is dependence of observations a curse of dimensionality?

→ Certainly not a curse; no ways to model otherwise for DS

Time series $\rightarrow$ Digital signal.

$\rightarrow$ Statistical digital signal processing (DSP)

→ Time series — set of observations, each one being recorded at a specified time. (Annual GDP of a country, sales figure etc)

→ A discrete time series — the set of time points at which observations are made is a discrete set. (All above including irregularly spaced data)

→ Continuous time series — observations are made continuously over some time interval. It is a theoretical concept (Roughly ECG graph)

↓
we sample the data

→ A discrete valued time series — is one which takes discrete values, i.e. finite value. (No. of accidents, no. of transactions)

Graphs

Observations:-

- ① Trend
- ② periodic / oscillation
- ③ Fluctuations
- ④ combinations

Any time series → Divide and conquer.
↓
make them separately

and combine.

Image → pixel intensity is changing x and y.
each point have different co-ordinates
as it is a matrix.

Objective of TSA

- Forecasting
- - Content - anything is going wrong, ECG, production process
- Understanding feature. - seasonality cycle, trends.

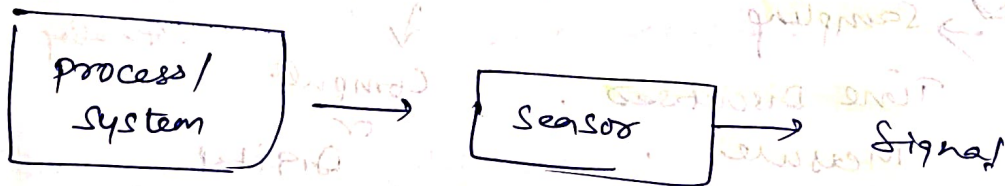
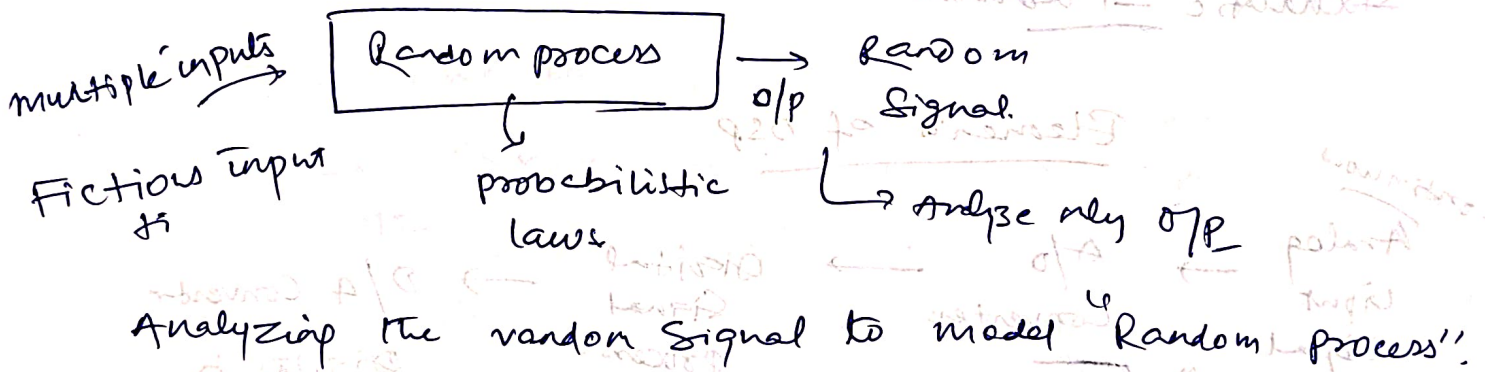
Trend Analysis :- predictability, seasonal variation

Correlation analysis :- Fluctuations and pattern, periodicity

Spectral Analysis :- Frequency Analysis.

Tasks in TSA

- ① Time series modelling
- ② Estimation - smoothing, filtering and predictions. Filtering in time
- ③ Data pre-processing - Detecting outlier



Ex: Heat Exchanger
Human Body
Atmosphere
Earth

usually h/w

• External or internal
ex. human body

Ex.

Temperature
• Blood pressure
• cloud movement
• seismic vibration
• Satellite image
• Stock price.

Signals

1D, 2D, 3D

↳ amplitude or value.

Dimensions

↳ independent variable or domain variable

value $\rightarrow$ Continuous or discrete

one or more of
time, space, frequency

Discrete-time (domain) signals $\rightarrow$ stem plots.

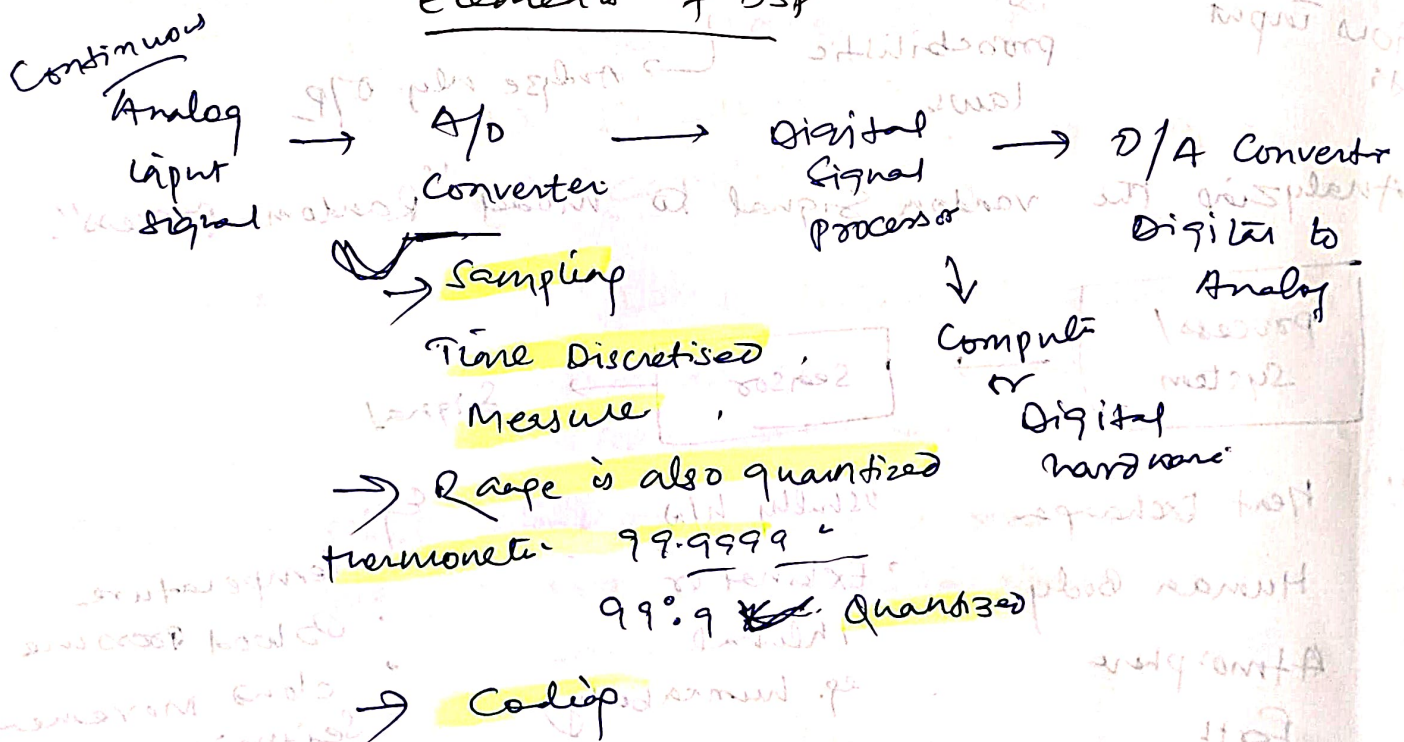
- if the sample size is large or the grid is too closely spaced then one generally connects one value to other using

Continuous line

Quantized / Discrete Digital $\rightarrow$ Discrete in time
Discrete in Amplitude

Stochastic $\rightarrow$ Random

Elements of DSP



$\rightarrow$ sampling doesn't result in loss of info

$\rightarrow$ quantization results $\rightarrow$ because rounding off

t_k — k^{th} sampling instant

$x\{t_k\}$ $k=1 \dots N$ is the discrete time signal

$$t_k = k T_s$$

Regular sample

T_s — Sampling period
0.15 ✓

$$x\{k_k\} = x[k T_s] \quad k=1 \dots N$$

$$f_s = \frac{1}{T_s} \quad \text{— sampling frequency}$$

Sampling Rate

$$x(k T_s) = x[k] \text{ or}$$

$x[n]$ — discrete time signals

T_s — time units/sample

$F_s \rightarrow$ sample/time

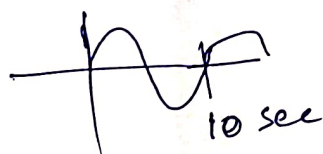
50 Hz

✓ 50 sample/second

$$\omega_s = 2\pi F_s$$

Periodic Sampling / Regular Sampling

$$x_a(t) = A \cos(\omega t + \theta)$$



Periodicity = 10

time seconds.

$F =$ cycle/sec

time
Period

no. of time units
seconds/cycles

$$\downarrow x_d[n] = A \cos(\quad)$$

$$x_a(t) = A \cos(\omega t + \theta)$$

↳ phase shift

↳ Continuous frequency

$$\omega = 2\pi F \rightarrow \text{cycles/time}$$

$$x_d[nT_s] = A \cos(\omega nT_s + \theta)$$

$$= A \cos[2\pi F nT_s + \theta]$$

$$= A \cos\left[2\pi \frac{F}{F_s} \cdot n + \theta\right]$$

$$f = \frac{F}{F_s}$$

Cycle
time
sample

time
cycle

$\frac{F_s}{F} \rightarrow f$

Discrete
frequency

Cycle
Sample

$$= A \cos(2\pi f + \theta)$$

$$= A \cos(\omega_n + \theta)$$

$$T_s = \frac{\text{time}}{\text{sample}}$$

$$F_s = \frac{\text{sample}}{\text{time}}$$

Q

$$F = 50 \text{ Hz}$$

$$f_{s,1} = 200$$

$$f_{s,2} = 40 \text{ samples/sec}$$

$$f_1 = \frac{F}{F_s} = \frac{50}{200} = \frac{1}{4}$$

$$f_2 = \frac{F}{F_s} = \frac{50}{40} = \frac{5}{4} = 1.25$$

Unique frequency range

$$0 \text{ to } 2\pi$$

$$-\pi < \omega < \pi \quad \text{--- Distinct}$$

$|\omega| > \pi$ --- alias $(+k\omega) \text{ mod } 2\pi = (\omega)$

$$|\omega| < \pi$$

Max frequency $\omega = \pi$

all higher

frequency
mapped

$$-\pi < \omega < \pi$$

$$\omega = 2\pi f$$

Range f

$$-1/2 < f < 1/2$$

$$-\pi < \omega < \pi$$

$$-1/2 < f < 1/2$$

$$-1/2 < \frac{F}{F_s} < 1/2$$

$$\boxed{F_s \geq 2F}$$

Nyquist Sampling
Criterion $\checkmark$

$$F_s = 4 \text{ samples/sec}$$

$$F_1 = 1 \text{ Hz}$$

$$f_1 = \frac{1}{4}$$

$$F_2 = 5 \text{ Hz}$$

$$f_2 = \frac{5}{4}$$

$$\omega = \frac{\pi}{4}$$

$$\left[\frac{\pi}{4} \right]$$

$$\omega = \frac{5\pi}{4}$$

$$\left[\pi + \frac{\pi}{4} \right]$$

≥ 3-4 times - in practice

anti-aliasing

Q. $x(t) = 3 \cos(100\pi t)$

Minimum
sampling
frequency.

3 cos

$$2\pi F = 100\pi$$

$$F = 50 \text{ Hz}$$

$$F_s = 2 \times 50 \text{ Hz}$$

$$= 100 \text{ Hz}$$

$$F = 50 \text{ Hz}$$

$$F_s = 200 \text{ Hz}$$

Q.2

$$f = \frac{F}{F_s}$$

$$F_s \geq 2F$$

Discrete
time signal

$$f = \frac{50}{200} = \frac{1}{4}$$

$$3 \cos(2\pi f n)$$

$$3 \cos\left(2\pi \frac{1}{4} n\right)$$

$$3 \cos\left(\frac{\pi n}{2}\right)$$

$$(c) F_s = 75 \text{ Hz}$$

$$3 \cos\left(\frac{50}{75} \times 2\pi n\right)$$

$$= 3 \cos\left(\frac{2}{3} \times 2\pi n\right)$$

$$3 \cos\left(\pi + \frac{\pi}{3}\right) n \quad \text{--- } 2\pi - 2\pi/3 \quad \checkmark$$

$$3 \cos\left(\frac{\pi}{3} n\right) \quad f = 1/3 \quad \checkmark$$

$$(d) \text{ Similar } f = \frac{F}{F_s} = \frac{50}{75}$$

$$3 \cos\left(\frac{\pi}{3} n\right)$$

$$\text{Continuous Version} \rightarrow f = \frac{F}{F_s} = 75$$

$$2\pi f = \pi/3 \quad \checkmark$$

$$f = \frac{1}{6} \quad \checkmark$$

$$\frac{1}{3} = \frac{F}{75}$$

$$F = 75/3 = 25 \text{ Hz} \quad \text{Alias cap} \quad \checkmark$$

Date
15/09/20

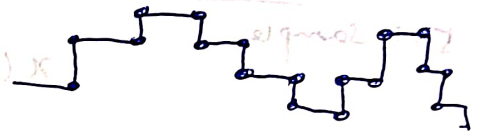
Reconstruction and Zero-order Hold

The inversion of the sampling operations, i.e. converting the sequence of numbers $x[n]$ to a continuous-time signal $x(t)$ is known as reconstruction.

Zero-order Hold (ZOH): - This is probably one of the most commonly used reconstruction devices.

St. line $\rightarrow$ One order

Parabola $\rightarrow$ Second order



Error $\rightarrow$ Slope

$$e_{ZOH} = \max |f(t_{k+1}) - f(t_k)| \leq h \max |f'(t)|$$

error - Maximum slope.

interval b/w two samples



h is small when f_s is large

Quantization - Rounding off

$$\text{Quantisation width} = \frac{(x_{\max} - x_{\min})}{(\text{No. of levels} - 1)}$$

Discretisation in Amplitude.

Analog to Digital

Discretisation in time

Quantization error = Rounding off error

Coding

number of bits

$$\geq \text{no. of levels}$$

(3)

7

9 levels

0V (min)

2V (max)

$$\text{Quantisation width} = \frac{x_{\max} - x_{\min}}{\text{No. of levels} - 1} = 0.25 \text{ V}$$

The possible quantised values are

0, 0.25, 0.5, 0.75, 1.00, 1.25, 1.50,

1.75, 2.00

$$x(5) = 1.66$$

5th sample

$$x(5) = 1.75 - \text{rounded to nearest level}$$

= 5th quantised level

$$2^{\text{no. of bits}} \geq \text{no. of levels}$$

Quantisation — Can be reduced by using more bits in codings thereby increasing the no. of quantisation level.

$$x = \{x[n]\}$$

$$x[n] = x_a(nT)$$

Delay (shift) of a sequence:

$$y[n] = x[n - n_0]$$

n_0 : Integer

Product of two sequence

$$x[n] \cdot y[n]$$

stretching

or

compressing the signal

Unit sample
Sequence

$$\delta[n] = \begin{cases} 0 & n \neq 0 \\ 1 & n = 0 \end{cases}$$

Any sequence is a weighted shifted version of basic unit signal.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

A sum of scaled, delayed impulses.

Basic Sequences

Unit step Sequence

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$u[n] = \sum_{k=-\infty}^n \delta[k]$$

$$\sum_{k=-\infty}^n \delta[k] = \begin{cases} 0, & \text{when } n < 0 \\ 1, & \text{when } n \geq 0 \end{cases}$$

Since $\delta[k] = \begin{cases} 0 & k \neq 0 \\ 1 & k = 0 \end{cases}$

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots$$

$$\delta[n] = u[n] - u[n-1]$$

$$\sum_{k=0}^{\infty} \delta[n-k]$$

First backward difference.

Exponential Sequence

$$x[n] = A \alpha^n$$

Eg: Trends

Increasing trends
Decreasing trends

Memory -
fades as time goes.

$$x[n] = A x^n u[n]$$

Simpler

unit sequence

If we want 0 for $t < 0$

Sinusoidal Sequence

$$x[n] = A \cos(\omega_0 n + \phi) \text{ for all } n$$

why it is important?

$2\pi f$ - fundamental frequency

Develop

Block Box

→ model

Experiments

- frequency limited to 0 to 2π
- Every signal ~~is rep~~ Can be represented in sine subject to condition

EIS - Electronic Impedance Spect

Frequency of a Discrete Signal - sample unit

~~$x[n]$~~

Q. What is the periodicity of the signal

$$x_4[n] = \cos(\pi n / 4)$$

$$x_1[n] = x_1[n+N]$$

$$\cos(\pi n / 4) = \cos(\pi(n+N) / 4)$$

$$\pi n / 4 + 2\pi k = \pi n / 4 + N\pi / 4 \quad k: \text{integer}$$

$$N = 2\pi k / \pi / 4 = 8k$$

$$k=1, \rightarrow \underline{\underline{N=8=2\pi/\omega_0}}$$

$$\frac{2\pi}{8} \rightarrow \frac{3\pi}{8}$$

$$x_1[n] = \cos(3\pi n/8)$$

$$x_1[n] = x_1[n+N]$$

$$\cos(3\pi n/8) = \cos[3\pi(n+N)/8]$$

$$\frac{3\pi n}{8} + 2\pi k = \frac{3\pi n}{8} + \frac{3\pi N}{8}, \quad k: \text{integer}$$

$$N = \frac{2\pi k}{\omega_0} = \frac{2\pi k}{(3\pi/8)}, \quad k=3, N=16$$

N is an integer

System / Filter:

System A system (or a process) is a set of units that process the input signal to produce an output.

→ A system is a filter and a filter is also a system

→ A filter is that unit which has been setup to process an input signal to produce a (desired) output.

? → Any system has its own frequency response. why amplify amplitude, so.

It behaves differently for different signals

Date
17/09/25

Lecture - 4 Extra class

— Dr. Anish
Monsleep

Simulation

Time series: $\rightarrow$ Measuring something continuous w.r.t time.

$\rightarrow$ Measuring something w.r.t time.

$\rightarrow$ Measuring something discrete w.r.t time

Process $\rightarrow$ Sensor $\rightarrow$ Signal

Spatial Signal : Temporal Signal

Image $\rightarrow$

temp, light, sound, n/w
wireless

Localized $\rightarrow$ time but changing across space.

Continuous signal: A signal that is defined for all time, i.e. it is continuous in both time and amplitude.

Discrete signal: A signal that is only defined at

eg: Specific time intervals. In this case, the continuous

working signal is sampled at discrete points in time.

$\rightarrow$ Sampled in time

Noise: Something stochastic in nature and that disrupts the original signal in a certain way

~~Deterministic~~

Deterministic Signal

A signal is deterministic if its behaviour or future value can be precisely determined from its past values or from a "Mathematical expression".

These signals are fully predictable at any point ~~of~~ in time.

Characteristics: No randomness involved. Can be described by a mathematical equation or rule.

Ex. Sine wave, Exponential Decay, A Sq. wave or a triangular wave.

Sampling

process of converting a continuous-time, continuous-amplitude signal into a discrete-time signal by taking samples at uniform intervals.

- eg: Ex: PM, groups in: Company

When sampling, if the sampling rate is not high enough (less than twice the highest frequency in the signal), aliasing occurs. This means that higher frequency components of the signal can be misrepresented as lower frequencies, leading to distortion.

Nyquist Theorem:- To avoid aliasing, the sampling rate should be at least twice the highest frequency present in the signal.

- A Continuous signal with frequency f Hz has to be sampled at least at twice this frequency

$$f_{s \text{ min}} = 2f$$

Date - 19/09/25

Lect - 4 Babji, Srinivasan

$$y[n] = x[2n] \rightarrow \text{stretching}$$

$$x y[n-1] = x[2(n-1)]$$

Causality

A system is causal if, for every choice of n_0 , the output sequence value at the index $n=n_0$ depends only on the input sequence value for $n \leq n_0$.

o/p depends only on past and present input (not on the future)

Forward difference system is not causal

$$y[n] = x[n+1] - x[n]$$

only Backward.

BIBO - Bounded-Input Bounded-Output Stability;

Input is bounded, output is also bounded.

$$\text{if } |x[n]| \leq B_x < \infty \text{ for all } n \\ \text{then } |y[n]| \leq B_y < \infty \text{, for all } n$$

But practically not possible to verify.

- Accumulator system is not stable.

input is constant

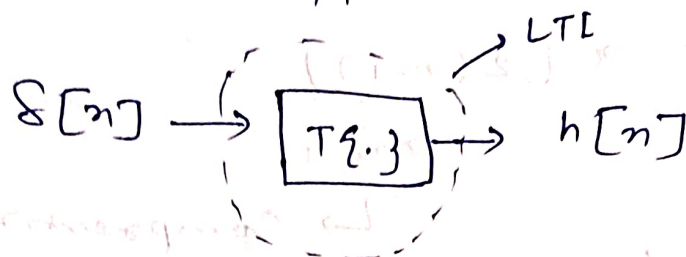
o/p - ramp

o/p - keeps on increasing unboundedly.

But it is easy to define, hard to verify.

Linear Time + Invariant (LTI) Systems

- Impulse response



$$y[n] = T \left\{ \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \right\}$$

Magnitude Unit Impulse

Convolution

Representation of general sequence as a linear combination of delayed impulse.

$x[n] * h[n]$ — Convolution.

Impulse response.

Weighted summation of input shifted impulses. responses

input
input shifted

→ Output of any LTI system, can be obtained using Convolution.

Integration → Continuous

Summation → Discrete

→ Weighted summation of impulse responses.

Q. $f(n) = [1 \ 2 \ 3 \ 4]$
 input

$h(n) = [1 \ -1]$
 impulse response

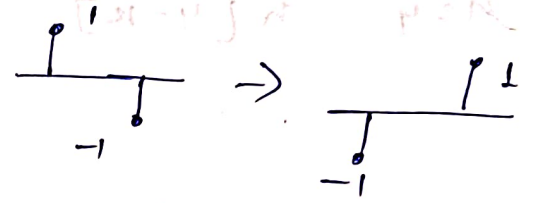
LTI system

~~$y[n] = x[n] * h[n]$~~

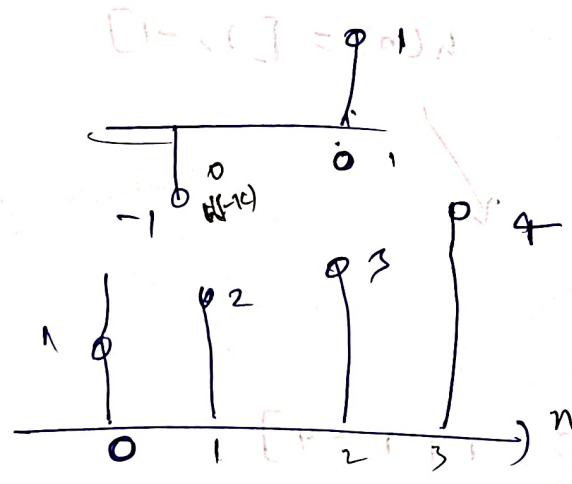
$y[n] = \sum x[k] h[n-k]$

$h(-k) = [-1 \ 1]$ ✓

$h(1-k) \rightarrow$



$n=0 \rightarrow x[k] h[-k] \rightarrow -1 \times 1 + 1 \times$
 $n=1$



$h(n) = [1, -1]$
 Time reversal $\rightarrow [-1, 1]$
 at $n=0$
 $n=0, -1 \times 1 + 1 \times 2 = -1 + 2 = 1$
 $n=1, -1 \times 1 + 1 \times 2 = -1 + 2 = 1$
 $n=2, -1 \times 2 + 1 \times 3 = -2 + 3 = 1$
 $n=3, -1 \times 3 + 1 \times 4 = -3 + 4 = 1$

at $n=2$

$n=3$
 $-1 \times 3 + 1 \times 4 = 1$

$-1 \times 2 + 1 \times 3 = 1$
 $n=4$
 $-1 \times 4 + 1 \times 5 = 1$

shifted unit
 $h[-k] = [-1, 1]$
 $h[1-k] = [-1, 1]$
 $h[2-k] = [-1, 1]$